BITS Pilani

Deep Neural Network
Module # 8: Attention Mechanism

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innovate achieve lead

The instructor is gratefully acknowledging
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materials freely available online.
This deck is prepared by Seetha Parameswaran.

Work Integrated Learning Programmes

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WHY STUDY ENCODER-DECODER?

Many NLP tasks require variable-length sequence-to-sequence mapping:

- Sentiment Analysis:
  - Input: "The phone battery lasts only 2 hours. Very disappointed."
  - Output: "Negative"
- Question Answering:
  - Input: Context + "What is the capital of India?"
  - Output: "New Delhi"
- Text Summarization:
  - Input: 500-word article
  - Output: 50-word summary

Key observation: Input length $T \neq$ Output length T' in general.

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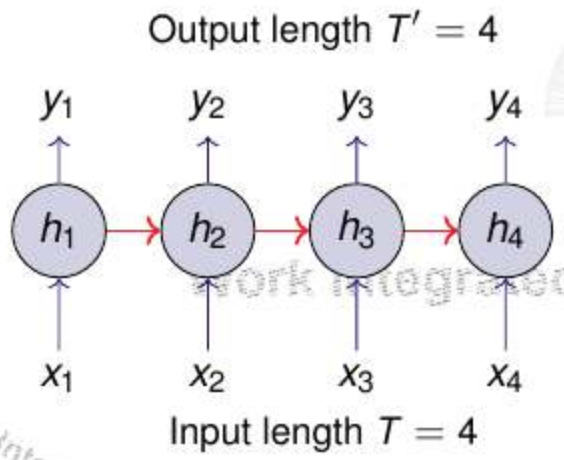
THE CHALLENGE: STANDARD RNN LIMITATIONS

Example: Question Answering fails with standard RNN

Standard RNN: Produces one output per input step (requires $T = T'$)



- Input: "India gained independence from Britain. What year was independence?" (10 words)
- Desired Output: "1947" (1 word)
- Problem: Standard RNN produces 10 outputs, not 1!

Solution needed: Separate **reading** phase from **generation** phase.

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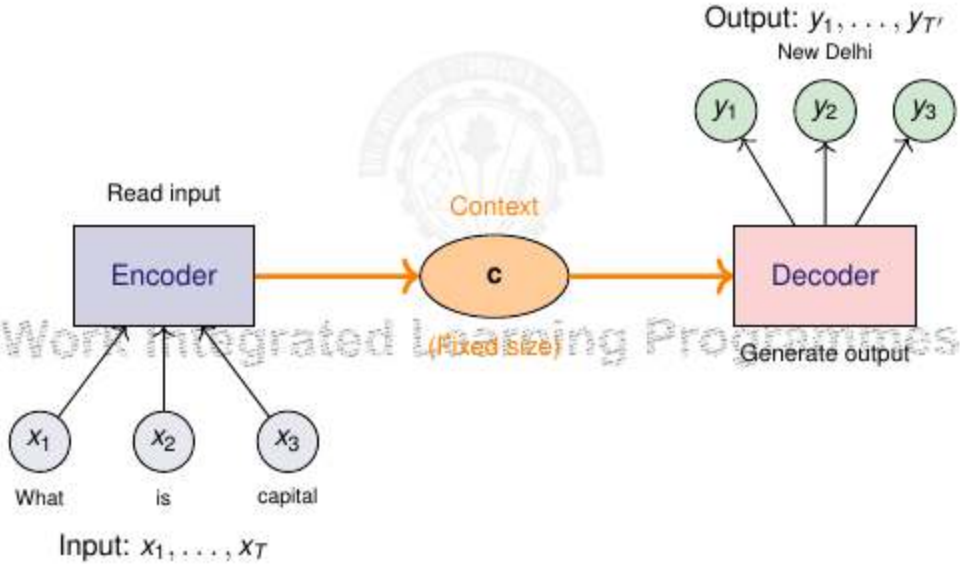
Solution: Encoder-Decoder Architecture

Key Idea: Two-stage process

1. Encoder: Read and compress entire input into context vector **c**
2. Decoder: Generate output sequence from context, one token at a time



Flexibility: T (input length) can differ from T' (output length).

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How Encoder-Decoder Works

- Phase 1: Encoding
  - Process input sequence with RNN:

$$h_t^{enc} = \phi(W_{hh}^{enc} h_{t-1}^{enc} + W_{hx}^{enc} x_t + b_h^{enc})$$

  - Extract context from final hidden state:

$$\mathbf{c} = h_T^{enc}$$

- Phase 2: Decoding
  - Initialize decoder with context:

$$h_0^{dec} = \mathbf{c}$$

  - Generate output sequence autoregressively:

$$h_t^{dec} = \phi(W_{hh}^{dec} h_{t-1}^{dec} + W_{hy}^{dec} y_{t-1} + b_h^{dec})$$

  - Output probability distribution:

$$P(y_t | y_{<t}, \mathbf{c}) = \text{softmax}(W_o h_t^{dec} + b_o)$$

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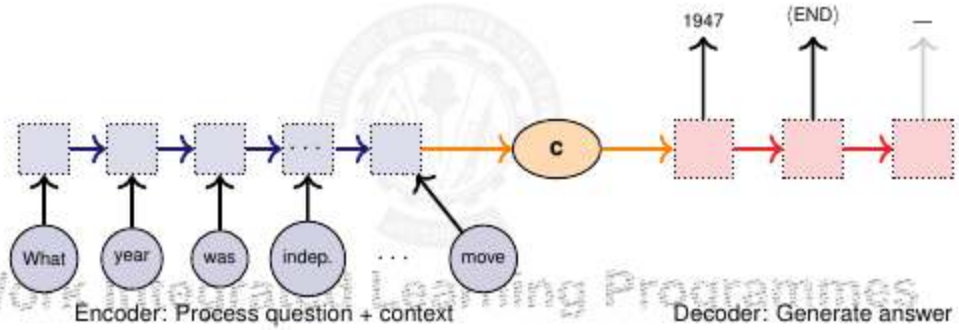
EXAMPLE: QUESTION ANSWERING WITH ENCODER-DECODER

- Task: Answer question from context

  Context: "India gained independence in 1947. Gandhi led the movement."

  Question: "What year was independence?"

  Answer: "1947"



- Process:
  - Encoder reads entire input ($T \approx 15$ tokens)
  - Context **c** encodes: "answer is about year and independence"
  - Decoder generates: "1947" then ⟨END⟩ ($T' = 1$ meaningful token)

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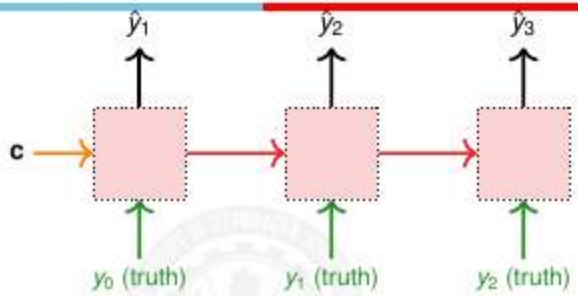
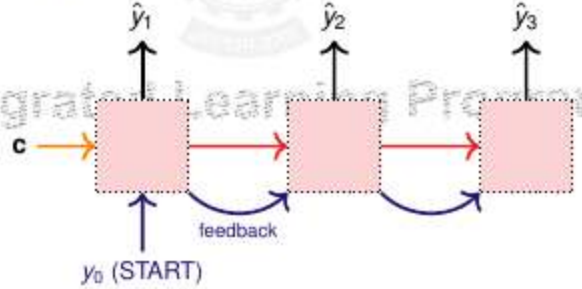
Training vs Inference

- Training (Teacher Forcing): Use ground truth as decoder input



- Inference (Free Running): Use model's own predictions



- Exposure Bias: Decoder trained on ground truth but tested on own predictions!

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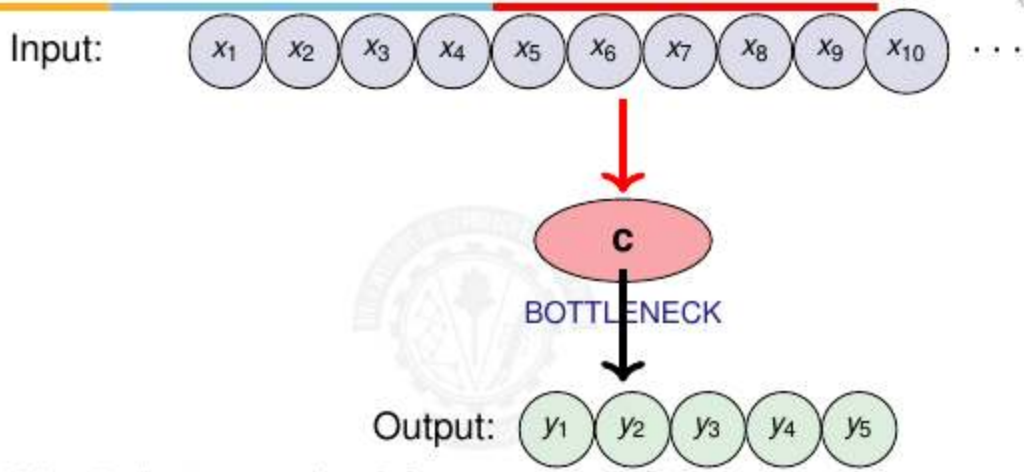
THE CRITICAL PROBLEM: CONTEXT BOTTLENECK

- All input information must pass through fixed-size vector **c**.



- Problems:
  1. Information loss: Long sequences → early tokens forgotten
  2. Fixed capacity: Vector size (e.g., 512-dim) limits information storage
  3. Uniform encoding: All input parts weighted equally, no selectivity
- Example: 50-word review compressed to 512 numbers, then generate sentiment!

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Bottleneck Impact: Concrete Example

- Task: Sentiment analysis of product review
- Input (30 words):

  "Ordered this laptop for work. Initial impressions were good. Fast delivery. However, after 2 weeks, battery drains in 2 hours. Screen flickers constantly. Very disappointed. Would not recommend."

Early positive sentiment lost?
Key phrases "very disappointed" weakened?

- Issue: When generating "Negative", decoder needs to emphasize later parts, but **c** treats all parts equally!

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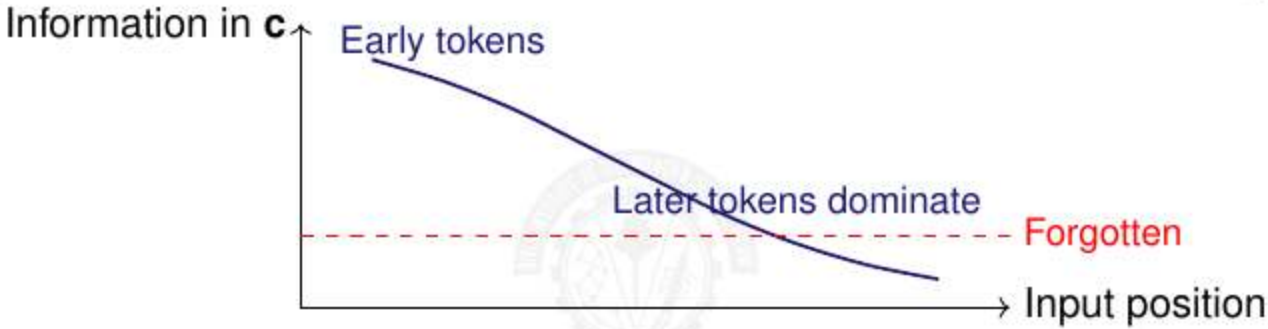
Why Fixed Context Fails for Long Sequences

Visualization of information loss:



Observations from research (Sutskever et al., 2014):

- Performance degrades significantly for sequences > 30 tokens
- First few words often lost by the time decoder generates output
- Model struggles with long-range dependencies

Question: How can we avoid this bottleneck?

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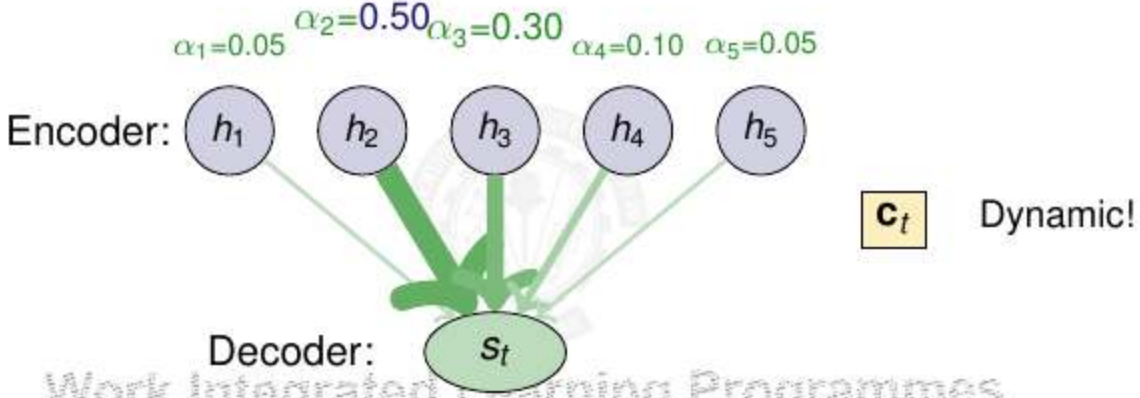
THE SOLUTION: ATTENTION MECHANISMS

- Core Insight: Instead of single fixed context, let decoder selectively attend to relevant input parts at each step.



- Key Differences from Encoder-Decoder:
  - Before: Single context $\mathbf{c}$ for entire decoding
  - After: Different context $\mathbf{c}_t$ at each decoder step t
  - Weights α_i learned automatically from data
  - Focuses on relevant input positions dynamically

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ATTENTION

Definition

Attention is a mechanism that computes a weighted sum of values based on the similarity between a query and keys.

- Components:
  - Query ($\mathbf{q}$): What we're looking for (decoder state)
  - Keys ($\{\mathbf{k}_i\}$): What's available to match (encoder states)
  - Values ($\{\mathbf{v}_i\}$): Actual content to retrieve (encoder states)
- Mathematical Form:

$$\text{Attention}(\mathbf{q}, \{\mathbf{k}_i, \mathbf{v}_i\}_{i=1}^{n}) = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$$

$$\text{Attention weights} \quad \alpha_i = \frac{\exp(\text{score}(\mathbf{q}, \mathbf{k}_i))}{\sum_{j=1}^{n} \exp(\text{score}(\mathbf{q}, \mathbf{k}_j))} \quad \text{(softmax normalization}$$

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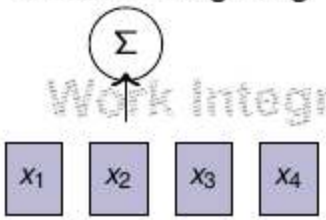
Core Concept: Weighted Combinations

Key Principle: Not all inputs contribute equally – attention learns relevance.

Without Attention:

$$\text{output} = f\left(\sum_{i=1}^{n} x_i\right)$$

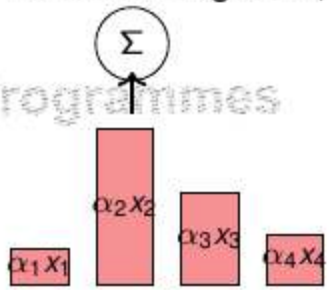
Uniform weighting



With Attention:

$$\text{output} = f\left(\sum_{i=1}^{n} \alpha_i x_i\right)$$

Learned weights α_i



Constraint: Weights sum to 1: $\sum_{i=1}^{n} \alpha_i = 1$ (probability distribution)

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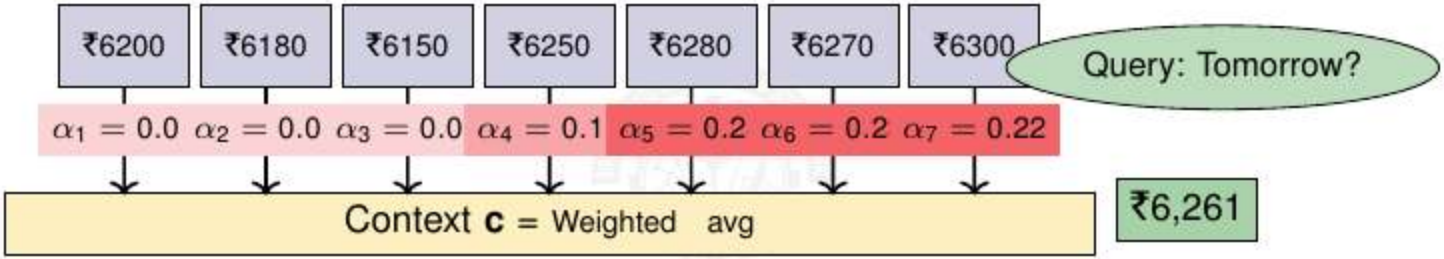
EXAMPLE 1: STOCK PRICE PREDICTION

Task: Predict tomorrow's gold price (₹/gram) using past 7 days.

Input: Gold prices = [₹6,200, ₹6,180, ₹6,150, ₹6,250, ₹6,280, ₹6,270, ₹6,300]



Observation: Recent days (5-7) receive higher attention weights α_i.

$$\mathbf{c} \approx 0.03 \times 6200 + 0.05 \times 6180 + 0.07 \times 6150 + 0.15 \times 6250 + 0.25 \times 6280 + 0.23 \times 6270 + 0.22 \times 6300$$
$$\approx 6261$$

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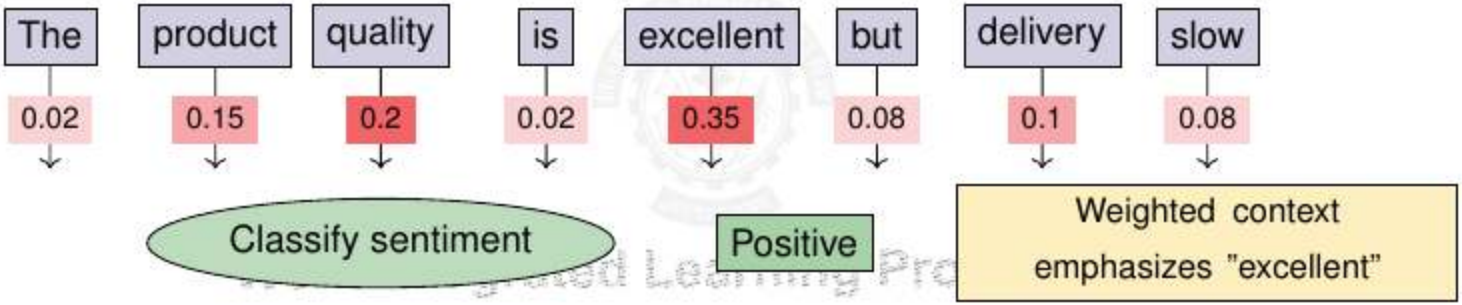
EXAMPLE 2: SENTIMENT ANALYSIS

Task: Classify review sentiment while identifying key phrases.

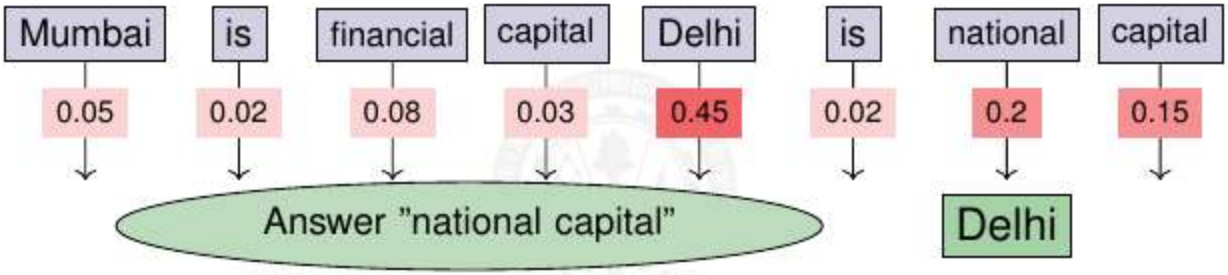
Input: "The product quality is excellent but delivery was very slow"



Key Insight: Model learns "excellent" ($\alpha_5 = 0.35$) is most important for sentiment, despite negative word "slow" present.

EXAMPLE 3: Q& A WITH ATTENTION

Context: "Mumbai is the financial capital. Delhi is the national capital."
Question: "What is the national capital?"



Attention weights show:

- Highest attention on "Delhi" ($\alpha_5 = 0.45$)
- Secondary attention on "national" ($\alpha_7 = 0.20$) and "capital" ($\alpha_4 = 0.15$)
- Model correctly ignores "financial capital"

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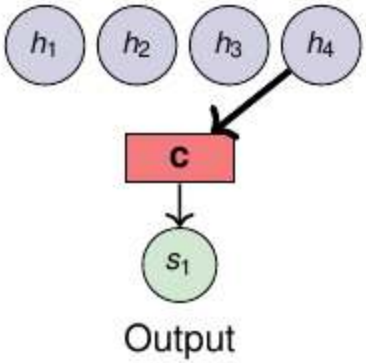
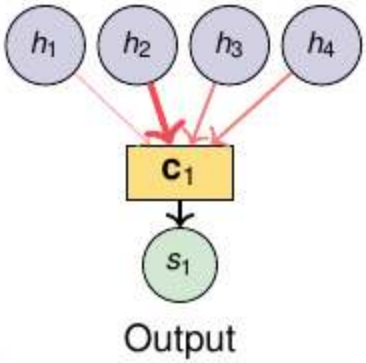
Without vs With Attention: Side-by-Side Comparison

Without Attention:



With Attention:



Issues:

- Fixed bottleneck **c**
- Information loss
- Poor long sequences

Benefits:

- Dynamic context $\mathbf{c}_t$
- Selective focus
- Handles long sequences

Key Difference: Context adapts to what decoder needs at each step!

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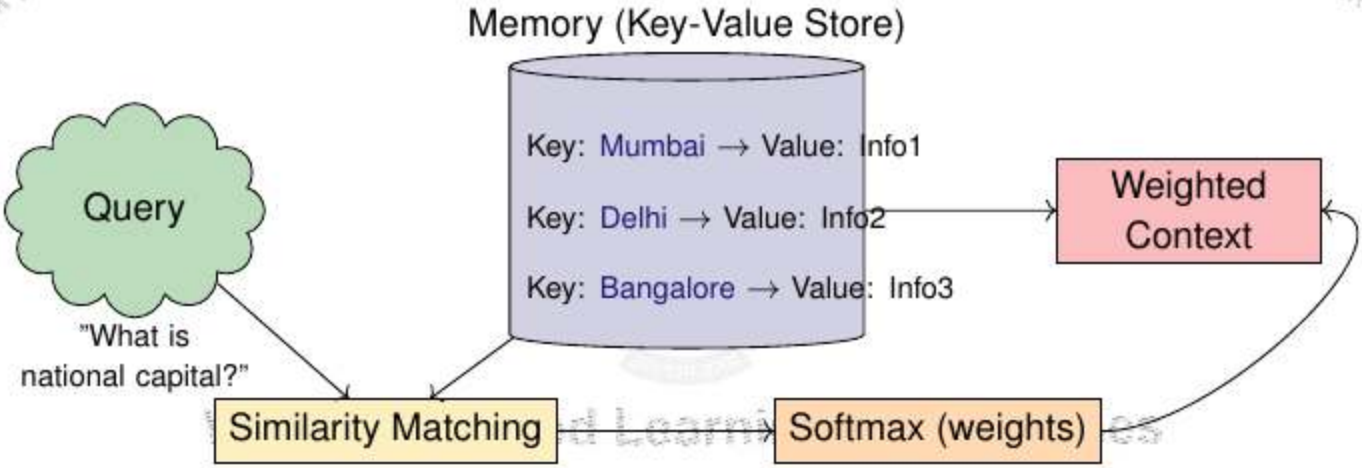
Attention as Information Retrieval

Analogy: Attention works like a database lookup system.



Process:

1. Query searches memory via **key similarity**
2. Compute attention weights based on query-key match
3. Retrieve **values** weighted by attention

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QUERY-KEY-VALUE (QKV): COMPONENTS

Three distinct roles in attention mechanism:

- Query ($\mathbf{q}$):
  - What information are we seeking?
  - Comes from decoder state: "I need to generate next word"
  - Example: Decoder state when generating "Delhi"
- Keys ($\{\mathbf{k}_i\}$):
  - What's available to match against?
  - Comes from encoder states: representations of input tokens
  - Example: Encoded representations of [Mumbai, is, financial, capital, Delhi, ...]
- Values ($\{\mathbf{v}_i\}$):
  - What actual content do we retrieve?
  - Also from encoder states (often same as keys)
  - Example: Contextual embeddings of input words

Intuition: Query searches Keys, retrieves matching Values.

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QKV Paradigm: Detailed Mechanics

Step-by-step attention computation:

1. Compute compatibility scores between query and all keys:

$$e_i = \text{score}(\mathbf{q}, \mathbf{k}_i) \quad \text{for } i = 1, \ldots, n$$

2. Normalize with softmax to get attention weights:

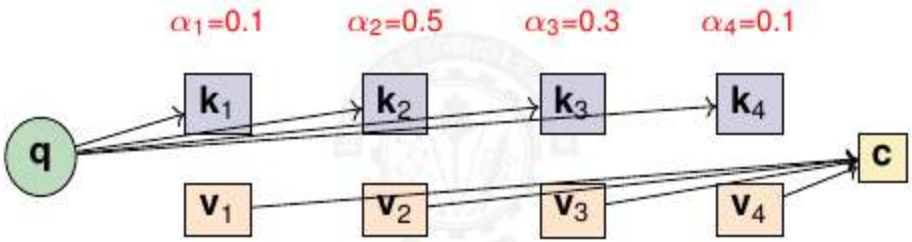
$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{n} \exp(e_j)} \quad \left(\text{ensures } \sum_i \alpha_i = 1\right)$$

3. Compute weighted sum of values:

$$\mathbf{c} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$$

QKV Paradigm: Visual

- Query $\mathbf{q}$: What information are we seeking? Decoder state
- Key $\mathbf{k}_i$: What's available to match against? Encoder state
- Value $\mathbf{v}_i$: What actual content do we retrieve? Encoder state



$$\alpha_i = \mathit{softmax}(\text{score}(\mathbf{q}, \mathbf{k}_i)) \quad \text{for } i = 1, \ldots, n$$

$$\mathbf{c} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$$

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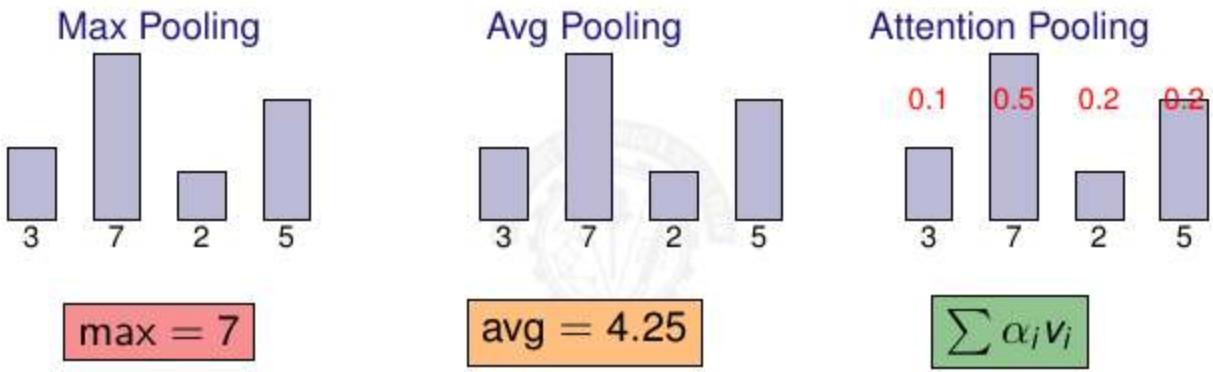
Attention as Weighted Pooling

Pooling Perspective: Attention is soft, adaptive pooling over sequence.



Key Difference:

- Max/Avg pooling: Fixed, uniform aggregation
- Attention: Data-dependent, learned aggregation based on query

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Dot-Product Attention

Simplest form: Measure similarity via inner product.

$$\text{score}(\mathbf{q}, \mathbf{k}_i) = \mathbf{q}^\top \mathbf{k}_i = \sum_{j=1}^{d} q_j k_{i,j}$$

$$\text{Attention } \alpha_i = \frac{\exp(\mathbf{q}^\top \mathbf{k}_i)}{\sum_{j=1}^{n} \exp(\mathbf{q}^\top \mathbf{k}_j)} = \text{softmax}(\mathbf{q}^\top \mathbf{K}^\top)_i$$

where $\mathbf{K} = [\mathbf{k}_1, \ldots, \mathbf{k}_n]^\top \in \mathbb{R}^{n \times d}$.

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EXAMPLE: DOT-PRODUCT ATTENTION

Example: Stock price prediction

- Query: $\mathbf{q} = [0.8, 0.6]^\top$ (tomorrow's expected trend features)
- Keys: $\mathbf{k}_1 = [0.5, 0.3]^\top$, $\mathbf{k}_2 = [0.9, 0.7]^\top$, $\mathbf{k}_3 = [0.4, 0.2]^\top$
- Scores:

$$
\begin{aligned}
e_1 &= 0.8(0.5) + 0.6(0.3) = 0.58 \\
e_2 &= 0.8(0.9) + 0.6(0.7) = 1.14 \quad \text{(highest!)} \\
e_3 &= 0.8(0.4) + 0.6(0.2) = 0.44
\end{aligned}
$$

- Weights: $\alpha_1 = 0.21$, $\alpha_2 = 0.62$, $\alpha_3 = 0.17$ (after softmax)
- Interpretation: Day 2 most aligned with query $\rightarrow$ highest attention.

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Scaled Dot-Product Attention

Problem: For large dimension d, dot products grow large $\rightarrow$ softmax saturates.
Solution (Vaswani et al., 2017): Scale by $\sqrt{d_k}$ to normalize variance.

$$\text{score}(\mathbf{q}, \mathbf{k}) = \frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

Benefits:

- Keeps scores in reasonable range regardless of dimension
- Prevents vanishing gradients in softmax
- Standard in all Transformer models

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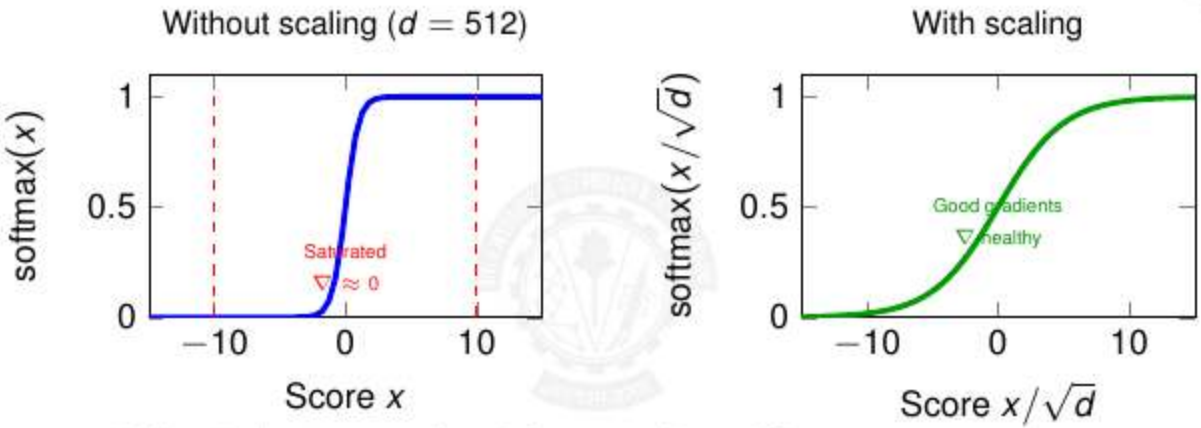
Why Scaling Matters: Gradient Flow

Effect of scaling on softmax gradients:



Impact:

- Left: Large scores push softmax to extremes (0 or 1)
- Right: Scaled scores maintain gradient flow
- Critical for training deep networks (6-24+ layers)

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Additive Attention (Bahdanau, 2014)

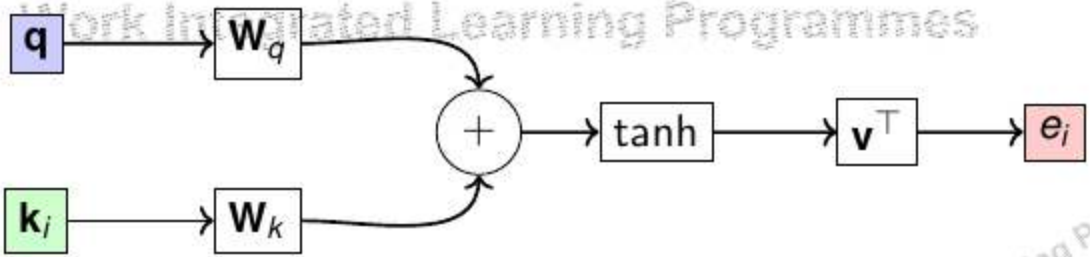
- Original attention mechanism for neural machine translation.

$$\text{score}(\mathbf{q}, \mathbf{k}_i) = \mathbf{v}^\top \tanh(\mathbf{W}_q \mathbf{q} + \mathbf{W}_k \mathbf{k}_i)$$

where:

- $\mathbf{W}_q \in \mathbb{R}^{h \times d_q}$: Query projection matrix
- $\mathbf{W}_k \in \mathbb{R}^{h \times d_k}$: Key projection matrix
- $\mathbf{v} \in \mathbb{R}^h$: Learnable weight vector
- h: Hidden dimension (hyperparameter, e.g., 256)

- Computation flow:



- Advantage: Can handle $d_q \neq d_k$ (different query/key dimensions).

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Comparison: Dot-Product vs Additive

Aspect	Dot-Product (Scaled)	Additive (Bahdanau)
Formula	$\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$	$\mathbf{v}^\top \tanh(\mathbf{W}_q\mathbf{q} + \mathbf{W}_k\mathbf{k})$
Parameters	None	$\mathbf{W}_q, \mathbf{W}_k, \mathbf{v}$
Complexity	$O(d)$ per pair	$O(d \cdot h + h)$ per pair
Expressiveness	Limited to linear	Nonlinear (tanh)
Parallelization	Excellent (matrix mult)	Moderate
Dimension req.	Requires $d_q = d_k$	Works with $d_q \neq d_k$
Modern usage	Transformers	Early seq2seq

Historical note: Additive was better for RNN-based models (2014-2016), but scaled dot-product dominates in Transformer era (2017+).

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Computational Complexity Analysis

For sequence length n, dimension d, and m queries:

Operation	Scaled Dot-Product	Additive
Score computation	$O(mnd)$	$O(mndh + mnh)$
Softmax	$O(mn)$	$O(mn)$
Weighted sum	$O(mnd)$	$O(mnd)$
Total	$\mathbf{O(mnd)}$	$\mathbf{O(mndh)}$

Matrix formulation (batch processing):

- Scaled dot-product: $\mathbf{QK}^\top \in \mathbb{R}^{m \times n}$ computed in one GPU operation
- Highly parallelizable on modern hardware (GPUs/TPUs)
- Memory efficient: $O(mn)$ for attention matrix

Practical impact: Dot-product is 3-5× faster in practice due to better hardware utilization.

Scoring Functions: Overview

The scoring function score($\mathbf{q}, \mathbf{k}$) measures query-key compatibility.

Method	Score Function	Parameters
Dot Product	$\mathbf{q}^\top \mathbf{k}$	None
Scaled Dot Product	$\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$	None
Additive (Bahdanau)	$\mathbf{v}^\top \tanh(\mathbf{W}_q \mathbf{q} + \mathbf{W}_k \mathbf{k})$	$\mathbf{W}_q, \mathbf{W}_k, \mathbf{v}$
Multiplicative (Luong)	$\mathbf{q}^\top \mathbf{W} \mathbf{k}$	$\mathbf{W}$

Design Trade-offs:

- Dot product: Fast (matrix ops), no parameters, requires $d_q = d_k$
- Scaled Dot Product : Prevents saturation for large dimensions (Transformer default)
- Additive: More expressive, flexible dimensions, more parameters

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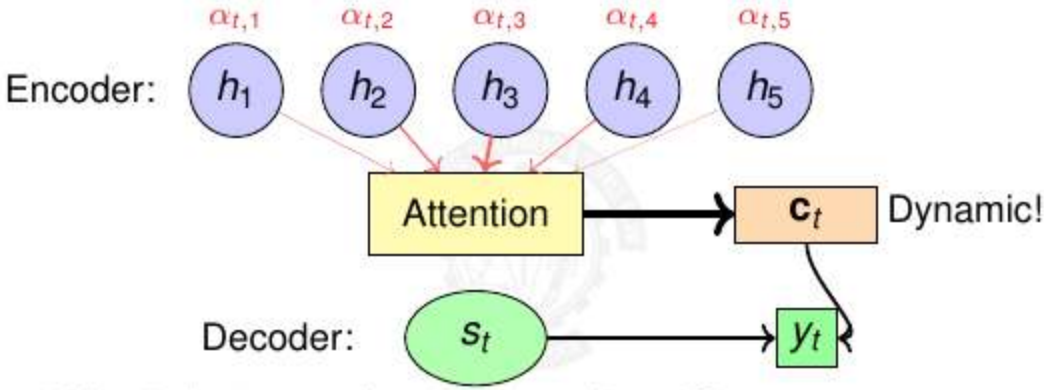
Bahdanau Attention: The Breakthrough

Key Innovation (Bahdanau et al., 2014): Replace fixed context with dynamic attention.



Context at time t:

$$\mathbf{c}_t = \sum_{i=1}^{T} \alpha_{t,i} \mathbf{h}_i$$

where $\alpha_{t,i}$ depends on current decoder state s_{t-1} and encoder state h_i.

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Bahdanau Attention: Math

At each decoder time step t:

- Step 1: Compute alignment scores
- Step 2: Normalize to attention weights
- Step 3: Compute context vector
- Step 4: Update decoder state
- Step 5: Generate output distribution

$$e_{t,i} = \mathbf{v}^\top \tanh(\mathbf{W}_s s_{t-1} + \mathbf{W}_h h_i)$$

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_{j=1}^{T} \exp(e_{t,j})}$$

$$\mathbf{c}_t = \sum_{i=1}^{T} \alpha_{t,i} \mathbf{h}_i$$

$$s_t = \text{RNN}([y_{t-1}; \mathbf{c}_t], s_{t-1})$$

$$P(y_t | y_{<t}, \mathbf{x}) = \text{softmax}(\mathbf{W}_o [s_t; \mathbf{c}_t])$$

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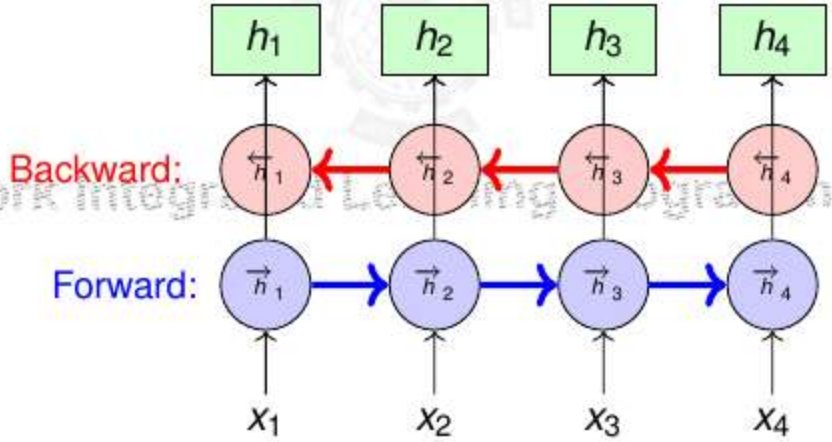
Encoder with Attention: Bidirectional RNN

Bahdanau used bidirectional RNN encoder to capture context from both directions.

$$\overrightarrow{h}_i = \text{RNN}_{\text{fwd}}(x_i, \overrightarrow{h}_{i-1})$$
$$\overleftarrow{h}_i = \text{RNN}_{\text{bwd}}(x_i, \overleftarrow{h}_{i+1})$$
$$h_i = [\overrightarrow{h}_i; \overleftarrow{h}_i]$$



Benefit: h_i encodes both past and future context for position i.

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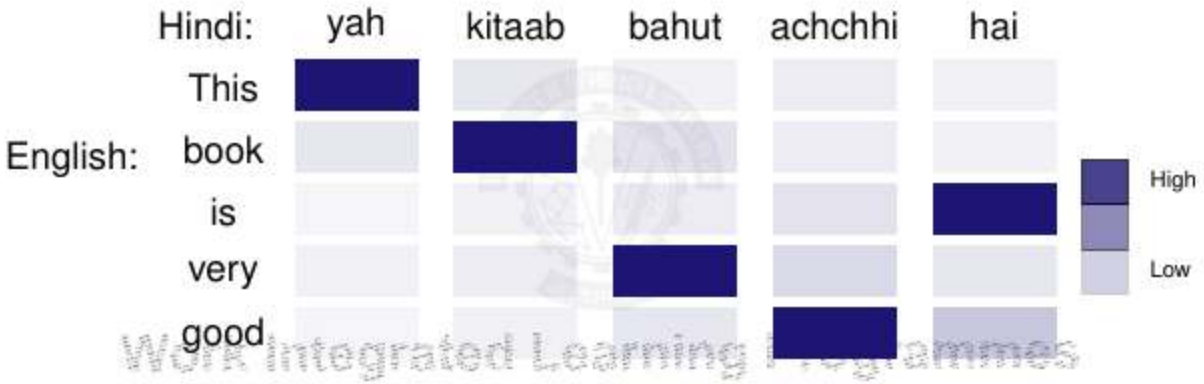
ATTENTION ALIGNMENT: VISUALIZATION

Example: Hindi to English translation

Source (Hindi): yah kitaab bahut achchhi hai

Target (English): This book is very good



Observations:

- Strong diagonal pattern: word-level alignment
- "yah" $\leftrightarrow$ "This": $\alpha = 0.85$
- "kitaab" $\leftrightarrow$ "book": $\alpha = 0.82$

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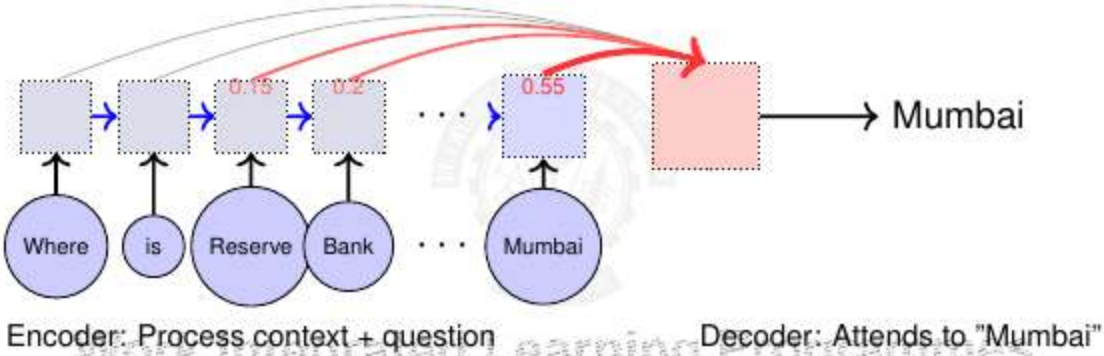
EXAMPLE: Q& A WITH ATTENTION

Context: "Mumbai is the financial capital. The Reserve Bank is located there."
Question: "Where is Reserve Bank?"
Answer: "Mumbai"



Attention interpretation:

- Decoder focuses on context mentioning "Reserve Bank" and "Mumbai"
- Highest weight ($\alpha = 0.55$) on "Mumbai" in context
- Successfully extracts answer from relevant part of input

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Implementation Considerations

- Training with Attention:
  - Teacher forcing: Use ground-truth y_{t-1} as decoder input
  - Loss function: Cross-entropy over target vocabulary

$$\mathcal{L} = -\sum_{t=1}^{T'} \log P(y_t^* | y_{<t}^*, \mathbf{x})$$

  - Optimizer: Adam with learning rate warmup and decay
  - Regularization: Dropout on encoder/decoder states and attention weights
- Inference with Attention:
  - Greedy decoding: Select $\arg\max_y P(y_t | y_{<t}, \mathbf{x})$ at each step
  - Beam search: Maintain top-k hypotheses (typical $k = 5$)
  - Attention recomputed: Fresh context $\mathbf{c}_t$ at each decoder step
  - Stopping: Generate until $\langle$END$\rangle$ token or max length
- Computational cost:
  - Attention: $O(T \times T')$ for input length T, output length T'

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Attention Types: Taxonomy

Type	Q Source	K,V Source	Use Case
Self-Attention	Same sequence	Same sequence	Intra-sequence relationships
Cross-Attention	Sequence A	Sequence B	Inter-sequence alignment (seq2seq)
Masked Attention	Same sequence	Same (causal mask)	Autoregressive generation
Multi-Head	Same (parallel)	Same (parallel)	Multiple relationship types

Key Distinctions:

- Source of Q, K, V: Where do query, key, value vectors come from?
- Masking: What positions can attend to what?
- Purpose: What relationships are being modeled?

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SELF-ATTENTION: DEFINITION AND PURPOSE

- Self-attention: Query, key, and value all come from the *same* sequence.

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V$$

  where $\mathbf{X} = [x_1, \ldots, x_n]^\top$ is the input sequence.
- Purpose: Model relationships *within* a sequence.
- Time Series Example: Stock price prediction using past 24 hours
  - Input: Gold prices over 24 hours: [₹6,180, ₹6,200, . . ., ₹6,280]
  - Self-attention allows each hour to attend to all other hours
  - Example: Hour 24 might attend strongly to hours 17, 10, 3 (7-hour periodicity)
  - Captures temporal patterns automatically
- Text Example: "The student didn't pass the exam because he was unprepared"
  - Self-attention helps "he" attend to "student"
  - Resolves coreference within the sentence
  - Captures long-range dependencies ("he" → "student")

Cross-Attention: Inter-Sequence Alignment

- Cross-attention: Query from one sequence, keys and values from *another*.

$$\mathbf{Q} = \mathbf{X}_A \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}_B \mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}_B \mathbf{W}_V$$

- Time Series Example: Multivariate forecasting
  - Sequence A: Stock prices (query current state)
  - Sequence B: Economic indicators (keys/values: inflation, GDP, interest rates)
  - Cross-attention: Stock prices attend to relevant economic indicators
  - Example: Gold price queries inflation rate and USD exchange rate
- Text Example: Hindi $\rightarrow$ English translation
  - Sequence A (Decoder): English words being generated
  - Sequence B (Encoder): Hindi source sentence
  - When generating "book", decoder attends to Hindi "kitaab"
  - Example: Generating "yah kitaab achchhi hai" $\rightarrow$ "This book is good"
    - ★ "book" (EN) attends to "kitaab" (HI) with $\alpha \approx 0.85$
    - ★ "good" (EN) attends to "achchhi" (HI) with $\alpha \approx 0.78$

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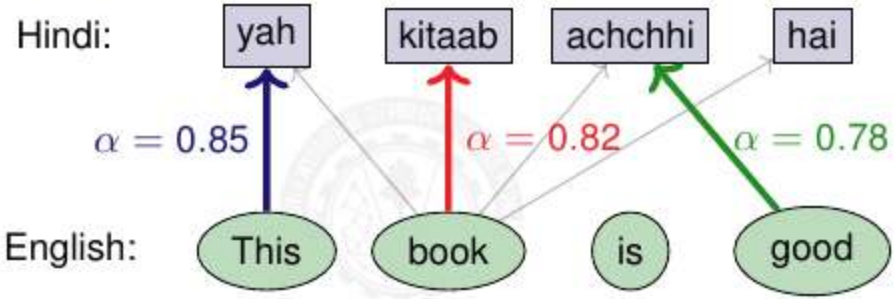
Cross-Attention: Visualization

- Machine Translation Example: Hindi to English



- Key: Decoder (English) dynamically attends to relevant parts of encoder (Hindi).
- This is exactly Bahdanau attention – the original cross-attention mechanism!

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MASKED ATTENTION: CAUSAL/AUTOREGRESSIVE

- Masked attention: Self-attention with *causal constraint* – can only attend to past.

$$\text{Mask}_{ij} = \begin{cases} 0 & \text{if } j \leq i \quad \text{(allowed)} \\ -\infty & \text{if } j > i \quad \text{(masked)} \end{cases}$$

- Time Series Example: Stock price prediction (prevent future leakage)
  - Predicting price at time t using only times $1, \ldots, t-1$
  - Mask prevents attending to future times $t+1, t+2, \ldots$
  - Ensures no "cheating" during training
  - Example: Predict tomorrow's gold price without seeing tomorrow's actual price
- Text Example: GPT-style language modeling
  - Task: Predict next word given previous words. "The customer was very ___"
  - Token at position 5 ("very") can only attend to positions 1-4
  - Cannot see future tokens during training or inference
  - Autoregressive: generate one token at a time, left-to-right

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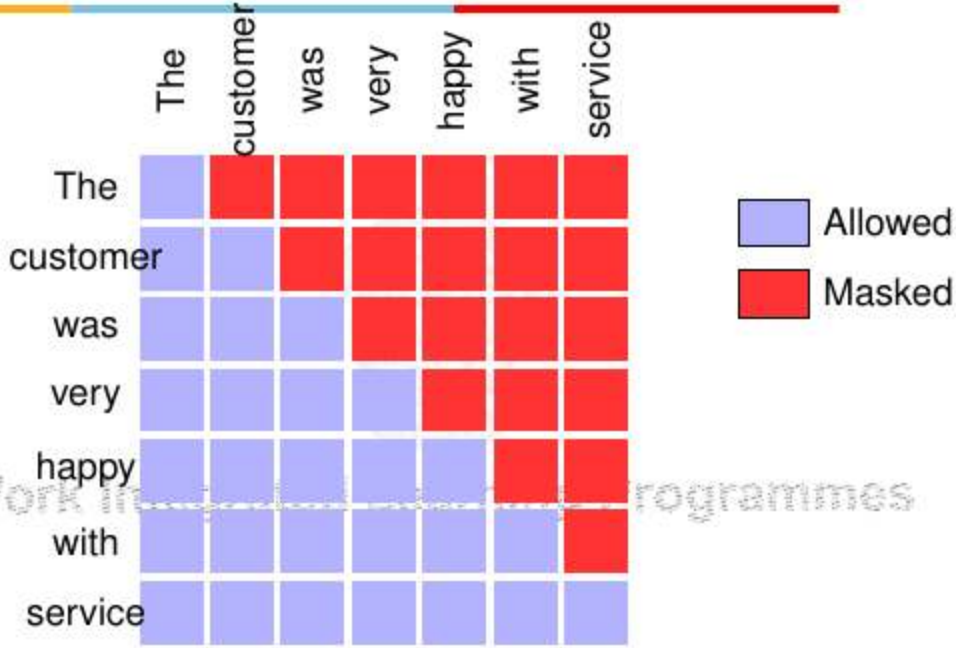
Masked Attention: Causal Mask Visualization

- Attention mask pattern: Query (row) attends to Key (column)



- Interpretation: Lower triangular pattern ensures token i only sees tokens $\leq i$.
- Implementation: Add $-\infty$ to masked positions before softmax $\rightarrow$ weight $= 0$.

Multi-Head Attention: Concept

- Motivation: Single attention may miss different types of relationships.
- Solution: Use multiple attention "heads" in parallel, each learning different patterns.
- Time Series Example: Gold price with 3 heads
  - Head 1: Attends to short-term trends (last 2-3 days)
  - Head 2: Attends to weekly patterns (7-day cycles)
  - Head 3: Attends to monthly patterns (30-day trends)
  - Combined: Richer representation than single attention
- Text Example: Sentence with 3 heads
  - Syntax head: Subject-verb-object relationships
    - "The customer complained" – verb attends to subject
  - Semantic head: Meaning and coreference
    - "he" attends to "customer" (resolves pronoun)
  - Positional head: Adjacent word dependencies
    - "very happy" – modifier-adjective pair

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Multi-Head Attention: Architecture

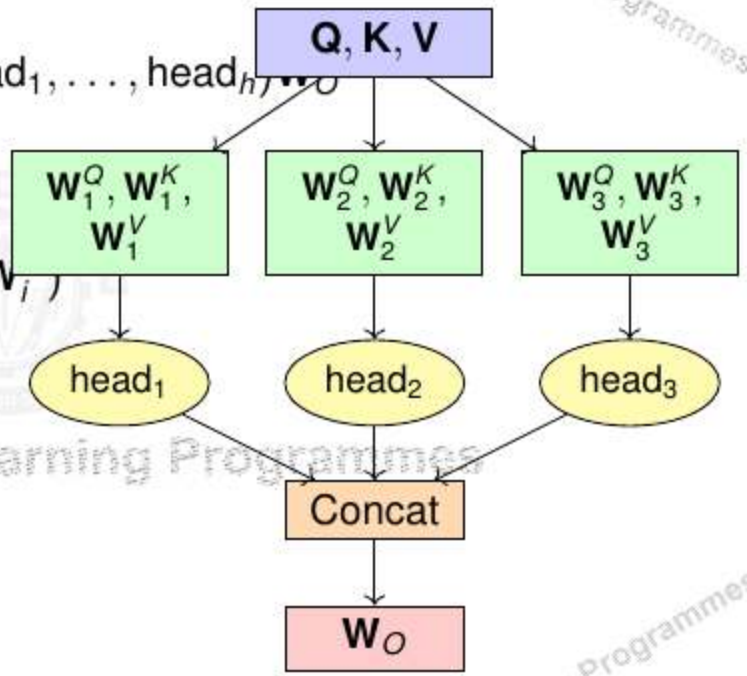
- Formula (Vaswani et al., 2017):

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}_O$$

where each head i computes:

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V)$$

- Typical hyperparameters:
  - $h = 8$ heads
  - $d_{\text{model}} = 512$
  - $d_k = d_v = 64$ per head



Multi-head output

When to Use Which Attention Type?

Attention Type	Use When...
Self-Attention	Modeling relationships within a single sequence Building contextual representations (e.g., BERT encoder) Capturing long-range dependencies in text or time series
Cross-Attention	Aligning two different sequences (seq2seq) Translation, summarization, question answering Transformer decoder attending to encoder outputs
Masked	Autoregressive generation (language modeling) Preventing information leakage from future GPT-style models, time series forecasting
Multi-Head	Capturing diverse relationship types simultaneously All modern Transformers (BERT, GPT, T5, ViT) When you need multiple attention patterns

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Self-Attention: Formal Definition

- Self-attention allows each position to attend to all positions in the *same* sequence.
- Given input $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{n \times d}$:
  1. Linear projections:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V$$

  2. Scaled dot-product attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

  3. Output:

$$\mathbf{Z} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{n \times d_v}$$

- Key property: Every output $\mathbf{z}_i$ is a weighted combination of *all* inputs $\{\mathbf{x}_j\}_{j=1}^n$.
- Difference from RNN: Direct connections between all positions (no sequential bottleneck).

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CNNs vs RNNs vs Self-Attention

Aspect	CNN	RNN	Self-Attn
Receptive field	Local (kernel k)	Sequential (history)	Global (all)
Path length	$O(\log n)$	$O(n)$	$O(1)$
Parallelization	High	Sequential	High
Complexity	$O(k \cdot n \cdot d^2)$	$O(n \cdot d^2)$	$O(n^2 \cdot d)$
Long-range	Many layers	Vanishing grad	Direct
Position info	Implicit	Built-in	Need encodin

- Key insight: Self-attention provides $O(1)$ path between any two positions!
- Trade-off: Better long-range modeling but $O(n^2)$ memory and computation.

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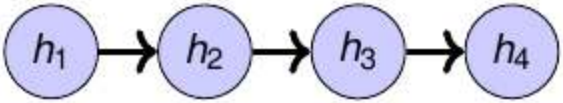
PARALLELIZATION: RNN VS SELF-ATTENTION

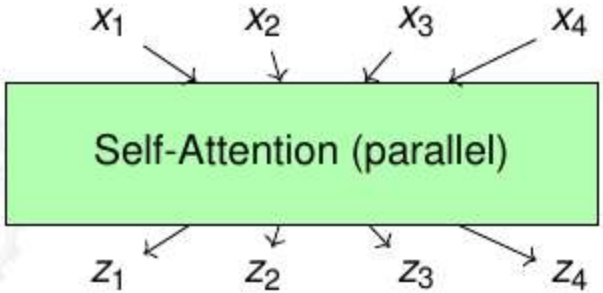
RNN (Sequential):



Must compute sequentially

Bottleneck: h_4 depends on h_3 depends on h_2 depends on h_1

Self-Attention (Parallel):



All positions in parallel

Benefit: All outputs computed simultaneously on GPU!

Result: Transformers train 10-100× faster than RNNs for long sequences.

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Position-Invariance Problem

- Critical issue: Self-attention is *permutation-invariant*!

$$\text{Attention}(\mathbf{X}) = \text{Attention}(\mathbf{PX})$$

  for any permutation matrix $\mathbf{P}$.
- Example: These sentences are treated identically:
  - Original: "The customer was happy with service"
  - Shuffled: "service with happy was customer The"
- Why? Attention weights depend only on *content*, not *position*.
- Problem for NLP: Word order matters!
  - "Dog bites man" $\neq$ "Man bites dog"
  - Need position information to distinguish
- Solution: Inject positional information into input embeddings.

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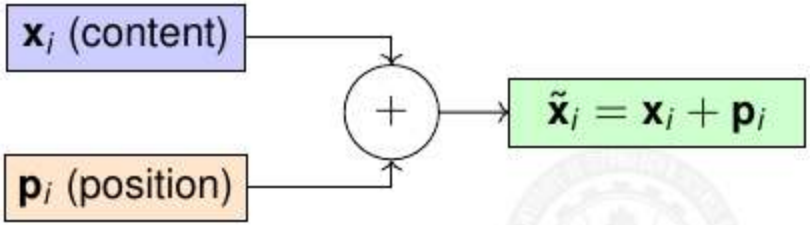
Positional Encoding: The Solution

Add position information to input embeddings:



Input to Transformer

Two approaches:

1. Learned positional embeddings (BERT, GPT-2):
   - $\mathbf{p}_i$ is a learned parameter for each position i
   - Simple, but doesn't generalize beyond training length
2. Fixed sinusoidal encoding (Original Transformer):
   - $\mathbf{p}_i$ computed using sine/cosine functions
   - Deterministic, generalizes to any sequence length

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Absolute Positional Encoding: Sinusoidal

Vaswani et al. (2017): Use fixed sinusoidal functions.

$$\mathrm{PE}(\mathrm{pos}, 2i) = \sin\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right)$$

$$\mathrm{PE}(\mathrm{pos}, 2i+1) = \cos\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right)$$

- pos: Position in sequence $(0, 1, 2, \ldots)$
- i: Dimension index $(0, 1, \ldots, d_{\mathrm{model}}/2 - 1)$
- d_{model}: Model dimension (e.g., 512)

Intuition:

- Different dimensions use different frequencies
- Lower dimensions = lower frequency (long-range patterns)
- Higher dimensions = higher frequency (fine-grained positions)
- Each position gets unique encoding vector

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Why Sinusoidal Encoding Works

- Key mathematical property: For any fixed offset k, $\mathrm{PE}(\mathrm{pos}+k)$ is a *linear function* of $\mathrm{PE}(\mathrm{pos})$.

$$\mathrm{PE}(\mathrm{pos}+k, 2i) = \mathbf{M}_k \cdot \mathrm{PE}(\mathrm{pos}, 2i) \quad \text{for some transformation matrix } \mathbf{M}_k$$

- Trigonometric identity:

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

- Example: For dimension i, let $\omega_i = 1/10000^{2i/d}$

$$\begin{aligned}\mathrm{PE}(\mathrm{pos}+k, 2i) &= \sin((\mathrm{pos}+k)\omega_i) \\ &= \sin(\mathrm{pos}\omega_i)\cos(k\omega_i) + \cos(\mathrm{pos}\omega_i)\sin(k\omega_i)\end{aligned}$$

This is a linear combination of $\mathrm{PE}(\mathrm{pos}, 2i)$ and $\mathrm{PE}(\mathrm{pos}, 2i+1)$.

- Benefit: Model can learn to attend by *relative position* (e.g., "3 positions back").

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Summary: Attention Mechanisms

1. Attention Mechanism
   - Dynamic, weighted aggregation based on relevance
   - Query-Key-Value framework for information retrieval
   - Solves bottleneck by attending selectively
2. Scoring Functions
   - Dot-product: simple, fast
   - Scaled dot-product: prevents saturation (Transformer default)
   - Additive: more expressive, slower
3. Attention Types
   - Self: intra-sequence, Cross: inter-sequence
   - Masked: causal, Multi-head: diverse patterns
4. Positional Encoding
   - Required for position-aware self-attention
   - Sinusoidal: fixed, generalizable

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Key Formulas to Remember

- 1. Attention (general):

$$\text{Attention}(\mathbf{q}, \{\mathbf{k}_i, \mathbf{v}_i\}) = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i \quad \text{where} \quad \alpha_i = \text{softmax}\left(\text{score}(\mathbf{q}, \mathbf{k}_i)\right)$$

- 2. Scaled Dot-Product Attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

- 3. Multi-Head Attention:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}_O$$

- 4. Positional Encoding:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right), \quad \text{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$

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Thank you :)
All the best :)

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